[illegible]

[illegible]

Abstract

[illegible] LLM [illegible]TF-IDF [illegible] LLM [illegible]” [illegible]” [illegible] uniform-100 [illegible] MV-Align [illegible]MV-Align[illegible]/[illegible] cross [illegible] ID–[illegible] ID [illegible] Amazon Beauty [illegible] Toys&Games [illegible]MV-Align [illegible] HR@10 [illegible] 6.06% [illegible] 6.92%[illegible]HR_new@10 [illegible] 2.04%[illegible]HR_few@10 [illegible] 5.22%[illegible] HR [illegible]MRR/NDCG[illegible] TF-IDF [illegible] LLM [illegible]

Keywords

[illegible], [illegible], [illegible], [illegible], [illegible], [illegible]

1 [illegible]

[illegible] SASRec [?] [illegible]/[illegible] TF-IDF [illegible] LLM [illegible]

[illegible] TF-IDF [illegible] Beauty [illegible] HR@10 [illegible] NDCG@10 [illegible] +27.5% [illegible] +40.9%[illegible] LLM [illegible]“[illegible]”[illegible]uniform-100 [illegible] uni100[illegible] [?][illegible]/[illegible]

MV-Align [illegible]“[illegible]”[illegible] cross [illegible]DCN-V2[illegible] ID×Text [illegible] ID [illegible]

[illegible]

- [illegible] MV-Align [illegible] SASRec [illegible] TF-IDF-only [illegible] LLM [illegible]
- [illegible] Amazon Beauty [illegible] Toys&Games [illegible]SENet[illegible]cross[illegible]/[illegible]
- [illegible] cross [illegible]MRR/NDCG[illegible]HR[illegible] HR–[illegible]

[illegible] MV-Align [illegible]

2 [illegible]

[illegible]. [illegible] Transformer [illegible] SASRec [?] [illegible] BERT [illegible] [?] [illegible]GRU4Rec [?] [illegible] RNN [illegible] SASRec [illegible]

[illegible]. [illegible] LLM [illegible] TF-IDF [illegible] word2vec [illegible]BERT[illegible]GPT[illegible]

[illegible]. Deep & Cross Network (DCN) [?] [illegible] DCN-V2 [?] [illegible] DNN [illegible] CTR [illegible] ID-[illegible]

[illegible]. [illegible] InfoNCE [?] [illegible]/[illegible] [?] [illegible]-ID [illegible]

[illegible]. [illegible] [?] [illegible]

3 [illegible]

[illegible]

- **RQ1**[illegible]vs. [illegible] L2 [illegible]/LLM [illegible]
- **RQ2**[illegible]**SENet** [illegible] SENet [illegible]/[illegible] SENet [illegible]
- **RQ3**[illegible] **vs.** [illegible] 256 [illegible] vs. [illegible] 4×64 [illegible] HR@K[illegible]NDCG@K[illegible]MRR@K[illegible] **new** [illegible] **few** [illegible]
- **RQ4**[illegible]**Cross** [illegible] **Align** [illegible]
  - [illegible] + cross + align[illegible]
  - [illegible] + cross[illegible] align[illegible]
  - [illegible] + align[illegible] cross[illegible]
  - [illegible] cross[illegible] align[illegible]

4 [illegible]

4.1 [illegible] SASRec

[illegible] $s_u = [i_1, \dots, i_T]$[illegible] L[illegible]SASRec [illegible] Transformer [illegible] s_u [illegible] $\mathbf{h}_u \in \mathbb{R}^d$[illegible] i [illegible] ID [illegible] $\mathbf{e}_i \in \mathbb{R}^d$[illegible] $\tilde{\mathbf{e}}_i$[illegible]

4.2 [illegible]

[illegible] TF-IDF [illegible] $\mathbf{t}_i^{(0)}$ [illegible] V [illegible] LLM [illegible] $\{\mathbf{t}_i^{(v)}\}_{v=1}^{V}$[illegible]/[illegible]/[illegible]/[illegible]SENet[illegible]

$$\mathbf{z}_i^{(v)} = \mathbf{W}_v \mathbf{t}_i^{(v)} \in \mathbb{R}^d, \tag{1}$$

$$\mathbf{r}_i^{(v)} = \mathbf{z}_i^{(v)} \odot \sigma(\mathrm{MLP}_v(\mathbf{z}_i^{(v)})). \tag{2}$$

[illegible] MV-Align [illegible] ℓ_2 [illegible]

$$\hat{\mathbf{r}}_i^{(v)} = \frac{\mathbf{r}_i^{(v)}}{\|\mathbf{r}_i^{(v)}\|_2}. \tag{3}$$

[illegible] $g_v = \sigma(\theta_v) \in (0,1)$[illegible] 0–1[illegible] TF-IDF [illegible] $\mathbb{R}^d$[illegible]

$$\mathbf{t}_i = \mathbf{P}\Big[\mathbf{t}_i^{(0)};\ g_1 \hat{\mathbf{r}}_i^{(1)};\ \dots;\ g_V \hat{\mathbf{r}}_i^{(V)}\Big]. \tag{4}$$

4.3 [illegible]

[illegible] DCN-V2 [?] [illegible] $\mathbf{x}_i = [\mathbf{e}_i; \alpha_i \mathbf{t}_i]$[illegible]

$$\mathbf{x}^{(l+1)} = \mathbf{x}^{(l)} + \mathbf{x}^{(0)} \odot (\mathbf{W}_l \mathbf{x}^{(l)} + \mathbf{b}_l), \tag{5}$$

[illegible] MLP [illegible] $\tilde{\mathbf{e}}_i$[illegible] α_i [illegible]

4.4 □□□□□□

□□□□ InfoNCE [?] □□□□□□□ ID □□□□□□□□□□□□□□ $\{i\}_{i=1}^{B}$□□□□ □□□ $\mathbf{S}_{ij} = \cos(\mathbf{e}_i, \hat{\mathbf{r}}_j^{(v)})$ □□□ τ□□□□□□□□

$$\mathcal{L}_{\text{align}}^{(v)} = \frac{1}{B} \sum_{i=1}^{B} -\log \frac{\exp(\mathbf{S}_{ii}/\tau)}{\sum_{j=1}^{B} \exp(\mathbf{S}_{ij}/\tau)}. \tag{6}$$

□□□□□□ $\pi_v = \text{softmax}(\phi)_v$ □□□ s_{mv} □□□□□

$$\mathcal{L}_{\text{mv-align}} = s_{\text{mv}} \sum_{v=1}^{V} \pi_v \mathcal{L}_{\text{align}}^{(v)}. \tag{7}$$

□□□□□. □□□□□□□□□□□□□□□□□□□□□ $\text{pop}(i)$ □□□□□□□□

$$w_i = 1 + \beta \cdot \max\left(0, \frac{T - \text{pop}(i)}{T}\right), \tag{8}$$

□□□□□□□□□□□□□□□□ $\sum_i w_i$ □□□□□

4.5 □□□□

□□□□□

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda \cdot \mathcal{L}_{\text{mv-align}}, \tag{9}$$

□□ $\mathcal{L}_{\text{rec}}$ □□□□□□□□□□□□□□□□□□□□□□□□□□□□λ □□□□□□

4.6 □□□□□

□□□□□□□□□□□□□□□□□□□□□□□□□□ [?] □□□□□ $|I|$ □□□□□□L □□□□□ □□□□d □□□□□□B □□□□□□d_t □□ TF-IDF □□□□□d_l □□ LLM □□□□□V □□□□□□□d_v □□□□ LLM □□□C □□□□□□□

□□□□□□□. □ 1 □□□□□□□□□□□□□□□□□□□

□□□□. SASRec □□□□ $O(|I|d)$ □□□□□□$O(Ld)$ □□□□□□$O(d^2)$ □□ Transformer □□□□□□□□□ Q, K, V, O □□□□□ FFN □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□TF-IDF □□ $O(d_t d)$□□□□ LLM □□ $O((d_t + d_l)d)$□□□□□ $O(Vd_v d)$ □□□□□□□ SENet □□□□□ $O(d_t d)$ □□□□□□□□□□□□□□ $O(Cd^2)$□□□□ C □□□□□□□□□ 1□□□□□□□□□ $O(d^2)$ □□□

□□□□. Transformer □□□□□□□ $O(L^2 d)$ □□ $L \times L$ □□□□□□□□ $O(Ld^2)$ □□□□ FFN□□□□□□□ $O(|I|d)$ □□□□□□□ logits□□□□□□□□□ □□□□□□

- **TF-IDF**□□□□ $O(|I| \cdot d_t \cdot d)$ □□□□□□□□□□□□ $O(|I| \cdot d^2)$□
- □□□□□□□□□□□□□□□□□□□□□ $O(|I| \cdot (d_t + d_l) \cdot d)$□
- □□□□□□ SENet □□□ $O(V \cdot |I| \cdot d_v \cdot d)$ □□□□□ $O(|I| \cdot Vd \cdot d)$□

□□□□□□□□□□ $O(B^2 d)$ □□ InfoNCE □□□□□□□□□□ LLM □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ LLM □□□□

5 □□□□□□

□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

- **A** □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□$\text{lr}_{\text{text}} = 1 \times 10^{-3}$□□□ □□□□□□□□□□□□□□□□□□□ (λ, τ)□
- **B** □□□□□□□□□□□□□□□□□□□□□□□□□□□/DNN□□□□□□□□□□□

□□□□□□ *ID* □□□. □□□□□□□□□□□□□□□□□□□□□□ ID □□□□□□□□□ID □□□□□□□□□□□□□ A □□□□□□□□□□□□□□□□□ 0 □ epoch □□□

Algorithm 1 □□□□□A □□ + B □□□

Require: □□□ $\mathcal{D}$□□□ SASRec□□□□□□□ $\Lambda, \mathcal{T}$

1: **A** □□□□□□□□□□□ $(\lambda, \tau) \in \Lambda \times \mathcal{T}$ □□□□□□□□□□□□□□□□□□□□□□

2: **B** □□□□□□□ A □□□□□□□□□□□□□□□□□□□□□

---

6 □□

6.1 □□□□□

□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ $K \in \{5, 10, 20\}$ □□□□□□□□□□□□□□□□□□NDCG□MRR□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

- **new**□$[1, 3)$ □□□
- **few**□$[3, 10)$ □□□
- **frequent**□$[10, \infty)$ □□□

□□ uni100 □□□□□□□ A.1 □□□□□□□□□□□□□□ sanity check□□□□□□□□□□□□□□ leave-one-out □ next-item □□□□□□□□□□□ 1 □ ground-truth□□□□ HR@K□Hit@K □ Recall@K □□□□□□□□□□□□ HR@K □□□/□□□□□ MRR/NDCG □□□□□□□□□□□

6.2 □□

□□□□□□□□□□□□□□□□□□□□□□

- **SASRec** □□ **ID**□□□□/□□/□□□□□□□□□□run50epBase_stratified.sh□
- **SASRec + TF-IDF** □□□□ TF-IDF □□□□□□□□□□□□□□□□□□□□two_phase_run_tfi
- **SASRec + TF-IDF + LLM** □□□□□□□ TF-IDF □□□□□ LLM □□□Qwen2.5-7B□□□ SENet □□□□□□two_phase_run_tfidf_llm_stratified.s
- **MV-Align** □□□□□ TF-IDF □□□□ + 4 □□□□ LLM □□□□□□□/□□/□□/□□□□□□□ SENet□□□□□□□□□□□□□□□□two_phase_run_multiview_v2_stratifiec

□□□□□□□□□□□□□□□□ BERT4Rec [?]□S3-Rec [?]□□□□□□□□□□□□□□□□□□□

6.3 □□□□

□□□□□□□□□□□□□□ $d = 256$□□□□□□ $L = 50$□2 □ Transformer □ 2 □□□□□□ 512□Adam □□□□□□□ 10^{-4}□

□□□□□. □□□□ A □□□15 □ epoch□→ B □□□50 □ epoch□□A □□□□□□□□□ MRR@10 □□ (λ, τ)□□□□□□□□□ HR@10□□□□□□□□□□□$\text{lr}_{\text{text}} = 1 \times 10^{-3}$ □□□□□$\text{lr}_{\text{cross}} = 5 \times 10^{-4}$ □□□□/DNN □□$\text{lr}_{\text{backbone}} = \text{lr}_{\text{text}} \times 0.1$ □ B □□□

□□. □□□□□□□□□□□□□□□ $\lambda \in \{0.03\}$ □ $\tau \in \{0.1\}$□□□□□□□□□ $\lambda \in \{0.05\}$ □ $\tau \in \{0.05\}$□□□□□□□□□□□□□□

□□□□. □□□□□□ 2025□□□ $K \in \{5, 10, 20\}$ □□□□□□□□□□□□□□□□ new:[1,3)□few:[3,10)□frequent:[10,∞)□□□□□□□□□□□□□□ YAML □□□□□□□□□

□□□□. □ 2 □□□□□□□□□□□□□□□□□□□□□$|I| = 12,102$□$d = 256$□$d_t = 256$□$d_l = 1024$□$V = 4$□$d_v = 1024$□□□□□□□□□□□□□□□□□□□□□□ 6M □□□□□ LLM □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

7 □□□□

□ 3 □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ ID □□□□□□□□ TF-IDF □□□□□□□□□□ HR@10 □□ +27.5%□□□□ +10.6%□□□□□□□□

Table 1: [illegible]$|I|$[illegible]L[illegible]d[illegible]d_t[illegible]TF-IDF [illegible]d_l[illegible]LLM [illegible]V[illegible]d_v[illegible]

[illegible]	[illegible]	[illegible]
SASRec [illegible] ID[illegible]	$O(\|I\|d + Ld + d^2)$	$O(L^2d + Ld^2 + \|I\|d)$
SASRec + TF-IDF	$O(\|I\|d + Ld + d^2 + d_td)$	$O(L^2d + Ld^2 + \|I\|d_td + \|I\|d)$
SASRec + TF-IDF + LLM [illegible]	$O(\|I\|d + Ld + d^2 + (d_t+d_l)d)$	$O(L^2d + Ld^2 + \|I\|(d_t+d_l)d + \|I\|d)$
MV-Align [illegible]	$O(\|I\|d + Ld + d^2 + Vd_vd + d_td)$	$O(L^2d + Ld^2 + V\|I\|d_vd + \|I\|d)$

Table 2: [illegible]

[illegible]	[illegible]	[illegible]
SASRec [illegible] ID[illegible]	$\sim$3.5M	–
SASRec + TF-IDF	$\sim$4.5M	+1.0M
SASRec + TF-IDF + LLM [illegible]	$\sim$5.3M	+1.8M
MV-Align [illegible]$V=4$[illegible]	$\sim$11.5M	+8.0M

[illegible] *ID* [illegible]. [illegible] TF-IDF [illegible] HR@10 [illegible] 4.69% [illegible] 5.98%[illegible]+27.5%[illegible]NDCG@10 [illegible] 2.69% [illegible] 3.79%[illegible]+40.9%[illegible] TF-IDF [illegible] HR@10 [illegible] 5.97% [illegible] 6.60%[illegible]+10.6%[illegible]NDCG@10 [illegible] 3.32% [illegible] 4.41%[illegible]+32.8%[illegible]

[illegible]. [illegible]MV-Align $>$ TF-IDF+LLM $>$ TF-IDF $>$ [illegible] ID[illegible]MV-Align [illegible]14B[illegible] HR@10=5.95%[illegible] ID +26.9%[illegible]MRR@10=3.22%[illegible]NDCG@10=3.86%[illegible]MV-Align [illegible]7B[illegible] HR@10=6.92%[illegible] ID +15.9%[illegible]NDCG@10=4.51%[illegible] MRR@10=3.76%[illegible] **new** [illegible] HR_new@10 [illegible] 1.99%[illegible] ID [illegible] 1.91%[illegible]

[illegible]*new/few*[illegible]. [illegible] **new** [illegible] **few** [illegible]7B[illegible] HR_new@10 [illegible] 1.84%[illegible] 1.99%[illegible]+8.2%[illegible]HR_few@10 [illegible] 4.57% [illegible] 5.22%[illegible]+14.2%[illegible]14B[illegible]HR_new@10 [illegible] 1.63% [illegible] 1.82% [illegible]+11.7%[illegible]HR_few@10 [illegible] 3.39% [illegible] 3.75%[illegible]+10.6%[illegible]

7.1 [illegible]

[illegible] LLM [illegible]Qwen2.5[illegible]

[illegible]. [illegible] LLM [illegible]

`[TITLE] {text}`

[illegible] `{text}` [illegible]LLM [illegible]

[illegible]. [illegible]

(1) [illegible]/[illegible]: `Identify the item: [TITLE] {text}`
(2) [illegible]/[illegible]: `What are the main functions and features of [TITLE] {text}?`
(3) [illegible]: `Who is the target audience or user group for [TITLE] {text}?`
(4) [illegible]/[illegible]: `Categorize the item [TITLE] {text} and describe its context.`

[illegible] SVD [illegible] 64 [illegible] SENet [illegible]

8 [illegible]

[illegible] RQ1–RQ4[illegible] 7B Aggressive [illegible]

8.1 RQ2 & RQ4: SENet [illegible]

[illegible] 4 [illegible] SENet [illegible]/[illegible]

[illegible]. [illegible] HR@10[illegible] +8.4%[illegible] +7.7%[illegible]MRR@10 [illegible] 12.7%[illegible] 15.5%[illegible]NDCG@10 [illegible] 4.7% [illegible] 6.8%[illegible] **HR-**[illegible] Top-1 [illegible]

SENet [illegible]. [illegible] SENet [illegible] $<$1%[illegible] SENet [illegible] Cross [illegible]SENet [illegible] SENet [illegible] HR_new@10 [illegible] 3.4%[illegible]

[illegible]. [illegible] Cross [illegible] HR_new@10[illegible] +14.8%[illegible] +6.1%[illegible] HR [illegible] Top-1 [illegible]

8.2 RQ1: [illegible] vs [illegible] L2 [illegible]

[illegible] 5 [illegible]7B Aggressive[illegible]ZCA + L2[illegible] L2 [illegible]

[illegible]. [illegible] 5 [illegible]

- [illegible] HR@10 [illegible] 0.9%[illegible]MRR@10 [illegible] 1.6%[illegible]NDCG@10 [illegible] 1.4%[illegible]
- [illegible]HR +1.3%[illegible]NDCG +1.8%[illegible]MRR +1.1%[illegible]

[illegible]. [illegible] LLM [illegible] TF-IDF + LLM [illegible]——[illegible]

8.3 RQ3: [illegible] vs [illegible]

[illegible] 3 [illegible]256 [illegible]vs [illegible]4 × 64 [illegible] = 256 [illegible]

[illegible] *14B* [illegible]. [illegible] 14B[illegible]5.95% HR@10[illegible]5.79%[illegible]MRR 3.22% vs 3.20%[illegible]NDCG 3.86% vs 3.81%[illegible]

[illegible] *7B* [illegible]. [illegible] 7B[illegible]6.92%[illegible]6.66%[illegible] HR@10[illegible]MRR 3.76% vs 3.71%[illegible]NDCG 4.51% vs 4.41%[illegible]**few** [illegible] HR_few@10 [illegible] 4.57%[illegible] 5.22%[illegible] 7B[illegible]+14.2%[illegible]

[illegible]. [illegible] 14B[illegible] 7B[illegible]

8.4 [illegible]

[illegible]MRR/NDCG [illegible] 15.4% [illegible] 6.9%[illegible] HR[illegible]−7.7%[illegible]——[illegible] LLM [illegible]SENet [illegible] HR_few@10 +14.2%[illegible] HR_few@10 +10.6%[illegible]

8.5 [illegible] HR–[illegible]

HR [illegible]MRR/NDCG[illegible]——[illegible]$>$12K [illegible]

Table 3: [illegible]%[illegible]

[illegible]	[illegible]			new [illegible]			few [illegible]		
	HR@10	NDCG@10	MRR@10	HR@10	NDCG@10	MRR@10	HR@10	NDCG@10	MRR@10
[illegible]									
SASRec [illegible] ID[illegible]	4.69	2.69	2.07	1.71	1.01	0.79	3.33	1.98	1.56
SASRec + TF-IDF	5.69	3.77	3.18	1.68	1.27	1.11	3.32	2.36	1.99
SASRec + TF-IDF + LLM [illegible]7B[illegible]	5.79	3.81	3.20	1.63	1.19	1.00	3.39	2.39	2.00
MV-Align [illegible]14B[illegible]	**5.95**	**3.86**	**3.22**	**1.82**	**1.25**	**1.06**	**3.75**	**2.56**	**2.19**
[illegible]									
SASRec [illegible] ID[illegible]	5.97	3.32	2.49	1.91	1.10	0.84	4.61	2.53	1.87
SASRec + TF-IDF	6.60	4.41	3.73	1.83	1.33	1.17	4.76	3.24	2.78
SASRec + TF-IDF + LLM [illegible]7B[illegible]	6.66	4.41	3.71	1.84	1.39	1.24	4.57	3.10	2.64
MV-Align [illegible]7B[illegible]	**6.92**	**4.51**	**3.76**	**1.99**	**1.37**	**1.18**	**5.22**	**3.44**	**2.90**

Table 4: [illegible] 7B[illegible]Aggressive[illegible] SENet [illegible]%[illegible]

[illegible]	HR@10	NDCG@10	MRR@10	HR_new@10
[illegible]SENet + Cross[illegible]	6.92	4.51	3.76	1.99
− SENet	6.97 (+0.7%)	4.54 (+0.7%)	3.78 (+0.5%)	1.99 (0%)
− Cross	7.45 (+7.7%)	4.20 (−6.9%)	3.18 (−15.4%)	2.11 (+6.0%)
− SENet − Cross	7.43 (+7.4%)	4.20 (−6.9%)	3.19 (−15.2%)	2.08 (+4.5%)

Table 5: [illegible]7B[illegible]Aggressive[illegible]%[illegible]

[illegible]	HR@10	NDCG@10	MRR@10	HR_new@10
[illegible] 7B				
[illegible]	6.92	4.51	3.76	1.99
[illegible]	6.86 (−0.9%)	4.45 (−1.3%)	3.70 (−1.6%)	1.99 (0%)
[illegible]TF-IDF + LLM[illegible]				
[illegible]	6.66	4.41	3.71	1.84
[illegible]	6.75 (+1.3%)	4.49 (+1.8%)	3.75 (+1.1%)	1.84 (0%)

Table 6: [illegible]

[illegible]	**HR**	**MRR/NDCG**
[illegible]	↓[illegible] 7.7%[illegible]	↑↑[illegible]+15.4%[illegible]
SENet	≈[illegible]+0.7%[illegible]	≈[illegible]+0.5%[illegible]
[illegible]	↓[illegible] 0.9%[illegible]	↑[illegible]+1.6%[illegible]
[illegible]	↓[illegible] −1.3%[illegible]	↓[illegible] −1.1%[illegible]
[illegible]vs [illegible]	↑[illegible]	↑[illegible]

[illegible]. [illegible]" [illegible]" [illegible] Top-1 [illegible]MRR[illegible] Top-K [illegible] HR@K[illegible] Top-K[illegible] HR[illegible] Top-1 [illegible] MRR[illegible]

[illegible]. [illegible]

- [illegible]/[illegible] Top-K [illegible]
- [illegible]/[illegible] 1[illegible]
- [illegible] NDCG@K [illegible]

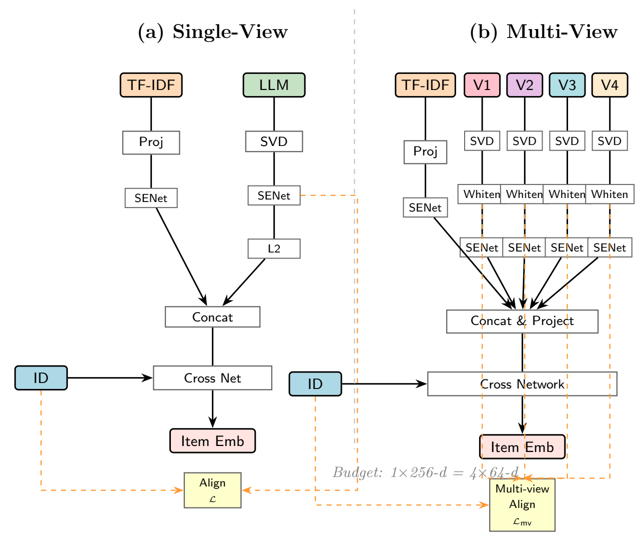

Figure 1: [illegible](a) [illegible] LLM [illegible] L2 [illegible](b) [illegible]InfoNCE [illegible]

9 [illegible]

[illegible] MV-Align [illegible]

(1) [illegible] TF-IDF [illegible] HR@10 +21.3%[illegible] +10.6%[illegible] ID [illegible]NDCG [illegible]+40.1% [illegible] +32.8%[illegible]

(2) [illegible] HR[illegible]+7.7%[illegible] MRR[illegible]−15.4%[illegible] NDCG[illegible]−6.9%[illegible] MRR −1.6%[illegible] 1-2%[illegible]

Table 7: vs sanity check

	Overall	Tail	
uni100 seed 1	—	—	—
uni100 seed 2	—	—	—

Table 8:

	Standard	Aggressive
λ	0.10	0.10
τ	0.05	0.05
cold_start_align_boost	2.0	2.5
inference_cold_text_boost	1.0	1.5
text_gate_init	0.7	0.7
text_gate_reg_l2	0.05	0.05

Table 9: Aggressive %

	HR@10	NDCG@10	HR_new@10	HR_few@10
MV-Align 7B	**6.92**	**4.51**	1.99	**5.22**
MV-Align 14B	6.84	4.43	1.96	5.19
MV-Align 32B	6.87	4.46	**2.04**	5.12

(3) MV-Align > TF-IDF+LLM > TF-IDF 14B HR@10=5.95% MRR@10=3.22% 7B HR@10=6.92% MRR@10=3.76% HR_new@10=1.99% 1.82%

(4) **HR-** HR

A

A.1 Sanity Check uni100 vs

[?] uni100 100

. checkpoint i ii uni100

. (1) sampled full Spearman/Pearson overall new/few/frequent (2) sampled full-ranking mis-ranking (3)

A.2

8 **Standard** Scale Law **Aggressive**

Scale Law *Standard*. Standard HR Scale Law

- HR@10: 32B (6.80%) > 14B (6.71%) > 7B (6.67%)
- HR_new@10: 32B (2.08%) > 14B (2.02%) > 7B (2.01%)
- NDCG_new@10: 7B/32B (1.45%) $\geq$ 14B (1.44%) $\approx$

Standard Scale Law

Aggressive . 9 Aggressive Scale Law 7B Scale Law

. Aggressive 7B 32B HR_new Standard Scale Law Aggressive

. $\lambda \in \{0.05, 0.10, 0.15\}$ $\tau \in \{0.03, 0.05, 0.10\}$ cold_boost $\in \{1.5, 2.0, 2.5\}$ HR@10 NDCG@10 HR_new@10

A.3 Center-only

i ii center-only overall new/few/frequent